Programming Exercise 1
Visual Analysis of Swift Parrot (*Lathamus discolor*) Observations in Australia

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1. Data Loading, Checking and Cleaning

The dataset ALA_S12026PE1.csv, comprising 25,276 records of Swift Parrot (Lathamus discolor) observations sourced from the Atlas of Living Australia (ALA, n.d.), was loaded into Tableau Public for initial visual exploration. The dataset spans records from 1900 to 2026 across eight Australian states and territories, drawn from 37 distinct data resources. Two irregularities were identified through visualisation and subsequently corrected using Python (pandas library). The resulting cleaned dataset, ALA_S12026PE1_cleaned.csv, contains 25,273 records and was reloaded into Tableau for all subsequent analysis.

1.1 Irregularity 1: Coordinate and State Province Errors

A scatter plot of decimalLatitude against decimalLongitude, coloured by stateProvince and filtered to exclude PRESERVED_SPECIMEN records, revealed two distinct location-based errors (Figure 1).
First, three HUMAN_OBSERVATION records carry a decimalLongitude of ~159.08°E, placing them approximately 500 km east of the Australian mainland in the Pacific Ocean — geographically implausible for Swift Parrot sightings, which are endemic to southeastern Australia (ALA, n.d.). All three are labelled as New South Wales, where typical longitudes average ~150.7°E. As correct coordinates could not be reliably inferred, these three records were removed, reducing the dataset to 25,273 records.
Second, four records carry incorrect stateProvince labels relative to their coordinates. One OBSERVATION record sourced from "Tasmanian Natural Values Atlas" is labelled New South Wales despite its coordinates (lat -42.08°, lon 148.42°) placing it firmly within Tasmania. Three records labelled "Australian Capital Territory" have coordinates at lon ~150.7–150.8°E, well outside ACT's eastern boundary of ~149.4°E and squarely within NSW. All four records were corrected in Python using their recordID — the NSW-labelled record was reassigned to Tasmania, and the three ACT-labelled records were reassigned to New South Wales.

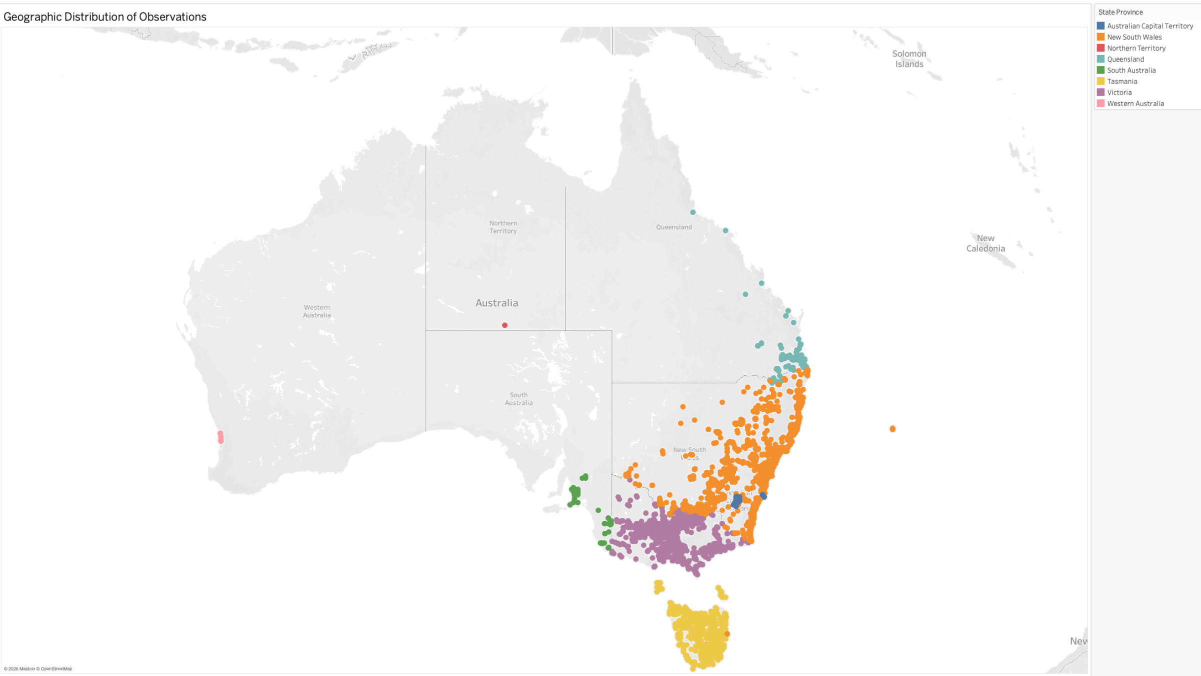

Figure 1. Geographic distribution of Swift Parrot observations (PRESERVED_SPECIMEN records excluded), revealing three records at longitude ~159°E outside Australia and four records with state labels inconsistent with their coordinates.

1.2 Irregularity 2: Data Resource UID–Name Mismatch

A highlight table plotting dataResourceUid against dataResourceName, with record counts encoded as colour intensity, revealed that UID dr1411 maps to two distinct resource names: "iNaturalist Australia"

(586 records) and "eBird Australia" (1 record) (Figure 2). Every other UID in the dataset maintains a strict one-to-one relationship with its resource name. The established UID for eBird Australia is dr2009 (9,894 records), confirming that the single record under dr1411 labelled "eBird Australia" is a metadata labelling error. The dataResourceName for this record was corrected to "iNaturalist Australia" in Python, restoring the expected one-to-one UID-to-name mapping across the dataset.

Data Resource UID vs Name

Data Resource Uid	Data Resource Name	
dr340	Australian Museum provider for OZCAM	35
dr341	Australian National Wildlife Collection provider for OZCAM	93
dr342	Museums Victoria provider for OZCAM	98
dr343	Northern Territory Museum and Art Gallery provider for OZCAM	1
dr344	Queensland Museum provider for OZCAM	5
dr345	Queen Victoria Museum Art Gallery provider for OZCAM	39
dr346	South Australian Museum Adelaide provider for OZCAM	26
dr347	Tasmanian Museum and Art Gallery provider for OZCAM	44
dr348	Western Australian Museum provider for OZCAM	5
dr359	BirdLife Australia, Birdata	2,151
dr360	Encyclopedia of Life Images - Flickr Group	3
dr364	ALA species sightings and OzAtlas	75
dr365	SA Fauna	25
dr368	NSW BioNet Atlas	3,300
dr466	Garden Bird Surveys	52
dr570	Historical Bird Atlas	518
dr571	First Bird Atlas	732
dr635	NatureShare	3
dr710	Tasmanian Natural Values Atlas	2,723
dr714	Gaia Guide	1
dr893	BowerBird	5
dr1010	ClimateWatch	106
dr1089	New South Wales Bird Atlassers	1,178
dr1097	Victorian Biodiversity Atlas	3,454
dr1132	WildNet - Queensland Wildlife Data	73
dr1411	eBird Australia	1
	iNaturalist Australia	586
dr1902	Earth Guardians Weekly Feed	4
dr2008	eBird Australia	1
dr2009	eBird Australia	9,894
dr2592	Logan City Council Species Sightings	2
dr22499	Museum of Comparative Zoology, Harvard University	11
dr22510	Observation.org, Nature data from around the World	17
dr29617	Xeno-canto - Bird sounds from around the world	9
dr30001	Academy of Natural Sciences of Philadelphia Ornithology Collection	1
dr30021	RBINS DaRWIN	3
dr30171	Naturalis Biodiversity Center (NL) - Aves	1
dr30608	Auckland Museum Land Vertebrates Collection	1

Figure 2. Data Resource UID vs Name consistency check. All UIDs map to a single resource name except dr1411, which incorrectly maps to both "iNaturalist Australia" and "eBird Australia".

2. Data Exploration and Visual Analysis

2.1 Question 1A: When Were These Birds Observed Over Time?

Two visualizations address the temporal distribution of observations: a line chart of annual counts (Figure 3) and a bar chart of monthly aggregates across all years (Figure 4). Records with null observationDate values (n = 3,403, ~13.5%) were excluded from both charts, acknowledged as a potential source of bias.

Figure 3 reveals a strongly right-skewed distribution. Annual counts remain below 50 from 1900 through the mid-1990s, reflecting the limited reach of early naturalist and museum-based programs. A clear inflection occurs around 2000, with a sharp and sustained rise from 2010 onward, peaking at 1,748 observations in 2024. This acceleration aligns with the mass adoption of citizen science platforms, particularly eBird Australia and iNaturalist Australia. The 2026 count (7 records) reflects partial-year data only.

Figure 4 shows a clear seasonal pattern, peaking in May (3,537 observations) with secondary peaks in April, June, and July — consistent with the Swift Parrot's known mainland wintering period (March–August). Lower counts in January–March and September–November align with the species' Tasmanian breeding season (BirdLife Australia, 2023).

A line chart is appropriate for the annual trend as it encodes continuity and direction of change over time. A bar chart suits the monthly distribution as months are discrete categories, with colour intensity applied as a secondary encoding to reinforce the seasonal peaks.

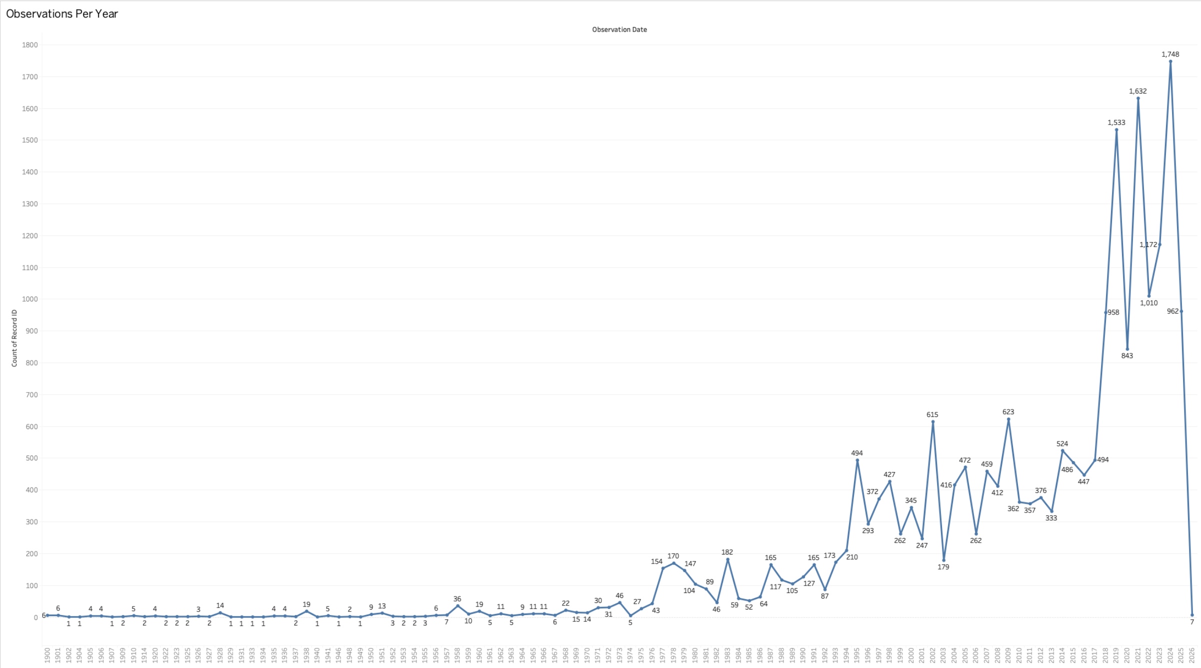

Figure 3. Annual Swift Parrot observation counts from 1900 to 2026, showing a marked increase from 2010 onward. Records with null observation dates are excluded (n = 3,403)

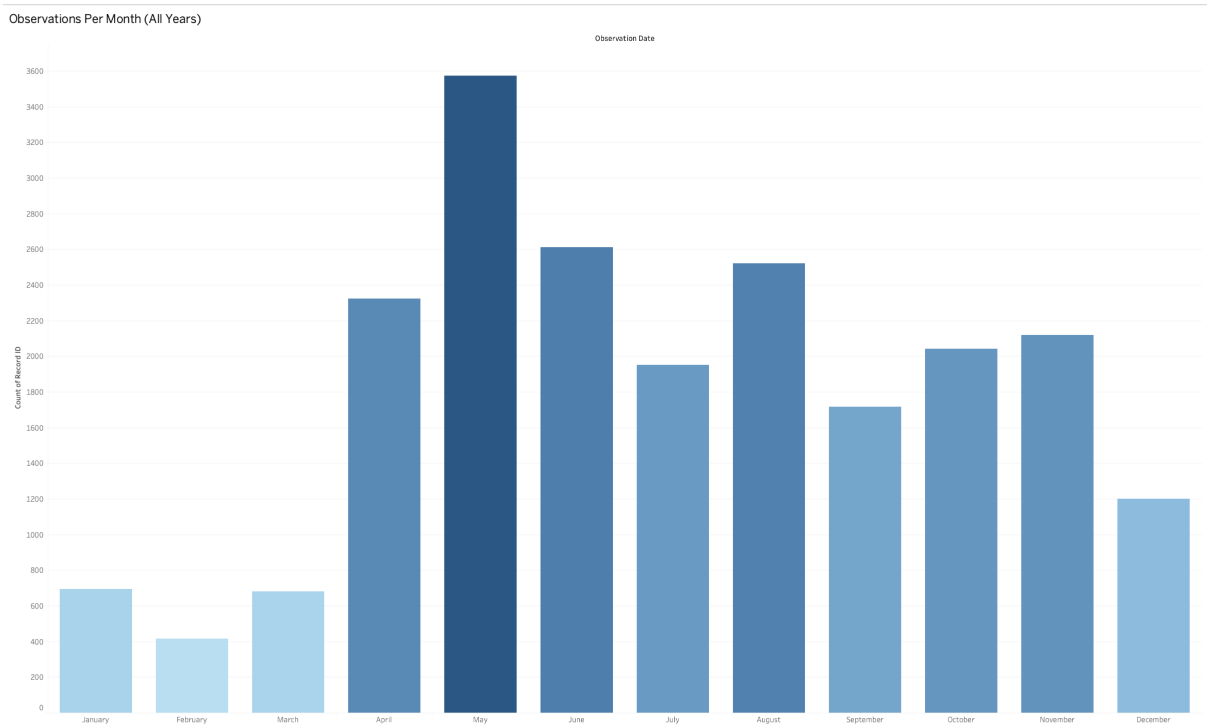

Figure 4. Monthly distribution of Swift Parrot observations aggregated across all years, with a pronounced peak in April–July consistent with the species' mainland wintering period.

2.2 Question 1B: How Have Observations from Different Data Resources Changed Over Time?

A stacked bar chart of annual record counts segmented by data resource was constructed, filtered to the top eight resources by total count to reduce visual clutter (Figure 5). Null observation dates were excluded as in Question 1A.

The chart reveals clear compositional shifts over time. eBird Australia (dr2009) contributes no records before ~2010 but rises sharply thereafter, dominating annual counts from 2015 onward — reflecting its rapid uptake within the Australian birding community. The Historical Bird Atlas and First Bird Atlas are concentrated between ~1977 and 1998 and cease contributing entirely afterward, consistent with their nature as time-bounded survey programs. The Tasmanian Natural Values Atlas and Victorian Biodiversity Atlas maintain stable moderate contributions from the mid-1980s onward, while NSW BioNet Atlas and BirdLife Australia Birdata emerge from ~2005.

A stacked bar chart is appropriate as it simultaneously encodes total annual volume through bar height and resource-level composition through colour segments — directly addressing the question's focus on change over time. Colour maps intuitively to the categorical distinction between resources.

A key concern is the dataset's heavy reliance on eBird Australia for recent records. Systematic biases in that platform's geographic coverage or observer effort distribution will disproportionately influence patterns observed across the entire analysis.

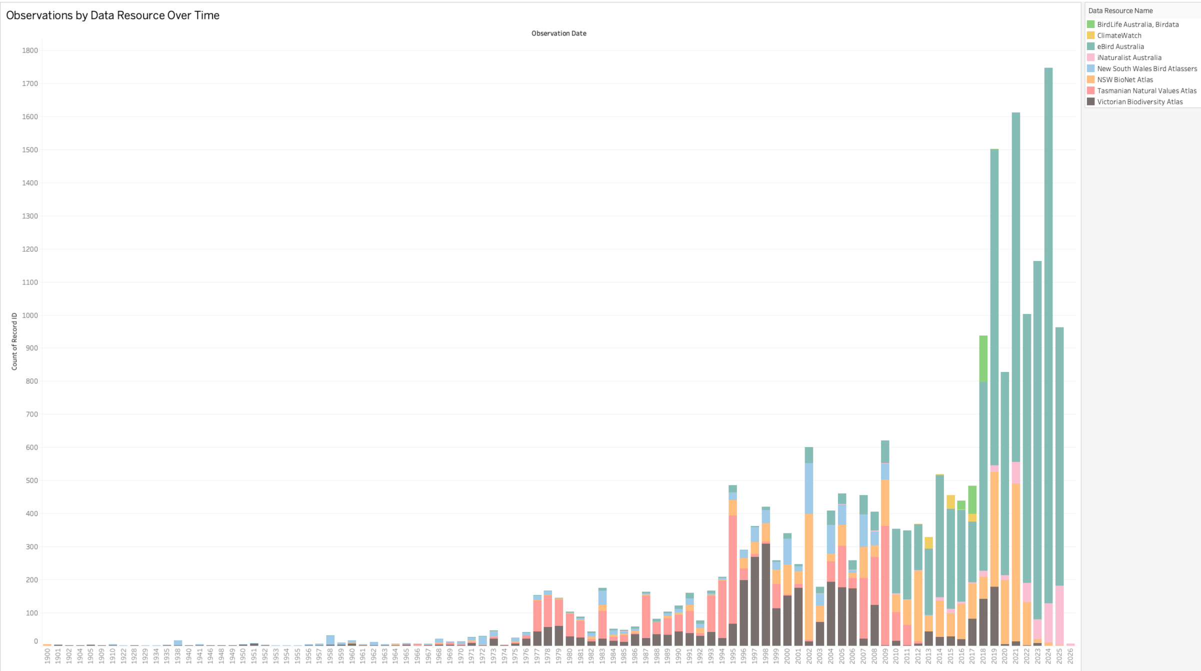

Figure 5. Annual Swift Parrot observations by data resource (top 8 resources shown by total record count), illustrating the emergence of eBird Australia as the dominant contributor from 2010 onward and the cessation of earlier atlas-based programs.

2.3 Question 1C: In Which Australian States and Territories Were These Birds Mainly Spotted?

Two complementary visualizations were produced: a sorted horizontal bar chart of counts by state (Figure 6) and a choropleth map of observation density by state (Figure 7).

Victoria records the highest count (8,768), followed by Tasmania (8,204) and New South Wales (6,683), collectively accounting for ~94% of all 25,273 cleaned records. The ACT (814) and Queensland (638) contribute modestly, while South Australia (158), Western Australia (6), and the Northern Territory (2) are negligible. This concentration in southeastern Australia is ecologically consistent with the Swift Parrot's known range, restricted to temperate eucalypt woodland in the southeast and Tasmania (ALA, n.d.).

The horizontal bar chart enables direct quantitative comparison across all eight states, with count labels providing precision and descending sort making rankings immediately apparent. The choropleth map complements this by grounding findings in geographic space, with colour intensity encoding density in a perceptually natural gradient — darker shading signifying higher counts.

A key caveat is that counts reflect both ecological range and observer effort distribution. Victoria, Tasmania, and NSW host the most active citizen science communities, which likely inflates their counts relative to a true effort-corrected representation.

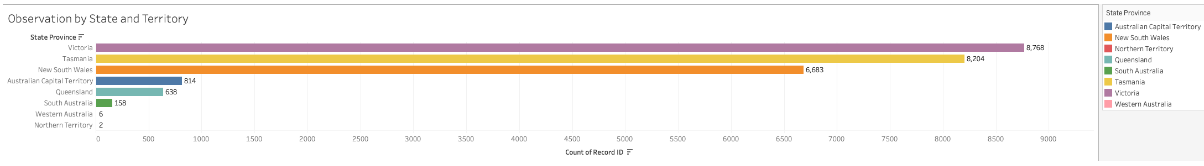

Figure 6. Swift Parrot observations by Australian state and territory. Victoria, Tasmania, and New South Wales collectively account for approximately 94% of all records in the cleaned dataset.

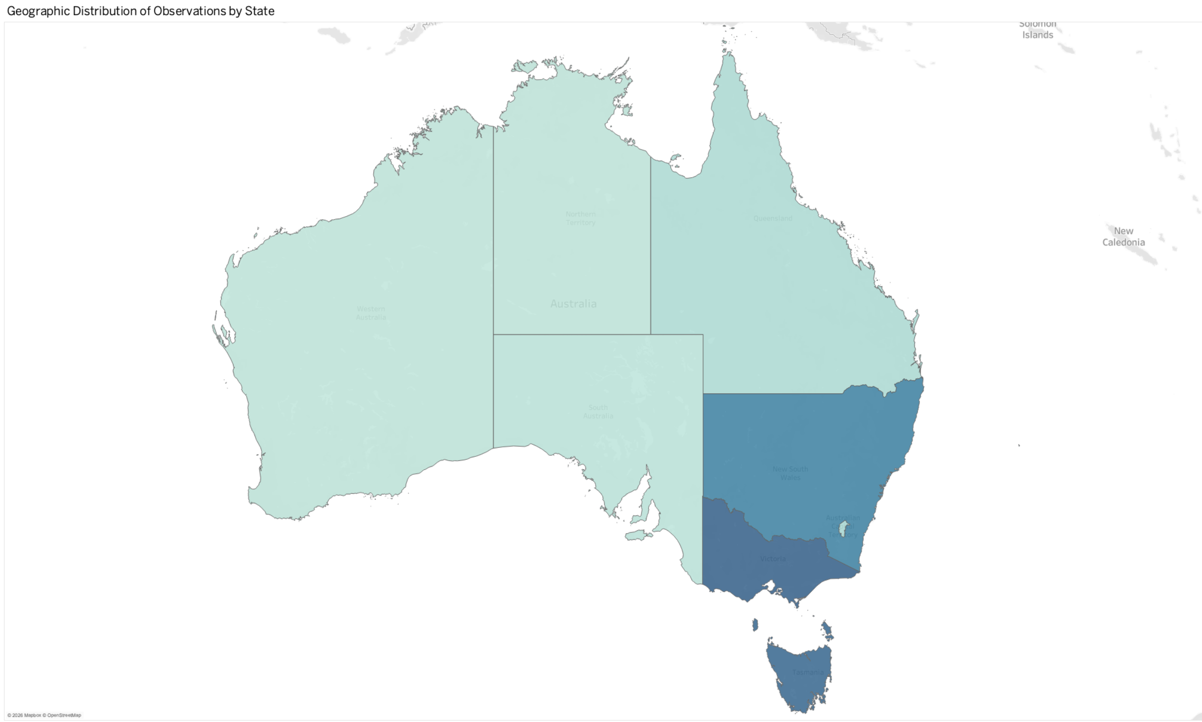

Figure 7. Choropleth map of Swift Parrot observation density by state, with darker shading indicating higher counts concentrated in southeastern Australia.

2.4 Question 2: Can We Determine Which Data Resources Best Show Seasonal Migration?

Based solely on the visualizations produced for Questions 1A–1C, it is not possible to definitively identify which data resources best capture the seasonal migration of the Swift Parrot.

To demonstrate seasonal migration at the resource level, one would need evidence of a given data resource consistently recording the species in Tasmania during its breeding season (September–January) and in mainland southeastern Australia during its wintering period (March–August). Establishing this requires a simultaneous cross-reference of three dimensions: data resource, month of observation, and state or territory.

The visualizations from 1A–1C each address one or two of these dimensions in isolation. Figure 4 confirms a clear aggregate seasonal pattern in the monthly distribution but does not disaggregate observations by data resource — it is therefore impossible to determine which resources drove observations in which months. Figure 5 shows how each resource's annual contribution has evolved over time, but operates at annual resolution, which is too coarse to detect within-year seasonal movements. Figures 6 and 7 show geographic concentration across states but contain no temporal or resource-level breakdown whatsoever.

Therefore, while the visual analysis strongly supports the existence of a seasonal pattern in the dataset as a whole — evidenced by the April–July mainland wintering peak in Figure 4 — determining which specific data resources best capture this migration pattern would require an additional visualization beyond the scope of the current analysis. A heatmap or faceted chart cross-referencing data resource, month, and state simultaneously would be necessary to answer this question definitively.

References

- ALA. (n.d.). *Atlas of Living Australia*. Retrieved February 27, 2026, from http://www.ala.org.au
- BirdLife Australia. (2023). *Swift Parrot Lathamus discolor*. Retrieved from https://www.birdlife.org.au/bird-profile/swift-parrot

Declaration of Generative AI Use

This assignment was completed with the assistance of Claude (Anthropic, claude.ai, Claude Sonnet 4.6), used in a guided capacity as permitted under Monash University's AI use guidelines. The tool was used for the following purposes:

- Identifying potential irregularities possibilities/ideas in the ALA dataset through programmatic analysis.
- Receiving guidance on Tableau interface for visualizing identified irregularities.
- Bug fixing and structuring Python code to correct coordinate errors and metadata mismatches in the dataset.
- Structuring written report content based on visualization findings.
- Condensing report sections to meet the page limit requirement.

All Tableau visualizations were independently built by me. Python code was reviewed and executed by the me. Analytical interpretations and conclusions are my own.